

# Audio Signals Are Periodic

Musical sounds are generated by vibrating systems

- Examples: **guitar strings**, **violin**, air columns (**flutes**)
- Most musical notes are approximately periodic signals

$$x(t + T) = x(t)$$

Because the signal is **periodic**, it can be represented using a **Fourier Series**



**Any periodic signal can be written as a sum of sinusoids**

$$x(t) = \sum_{k=1}^{\infty} A_k \sin(k\omega_0 t + \phi_k)$$

where

$A_k$  = amplitude of the  $k$ -th harmonic

$k\omega_0$  = harmonic frequency

$\omega_0 = 2\pi f_0$  = fundamental angular frequency

$f_0$  = pitch of the note

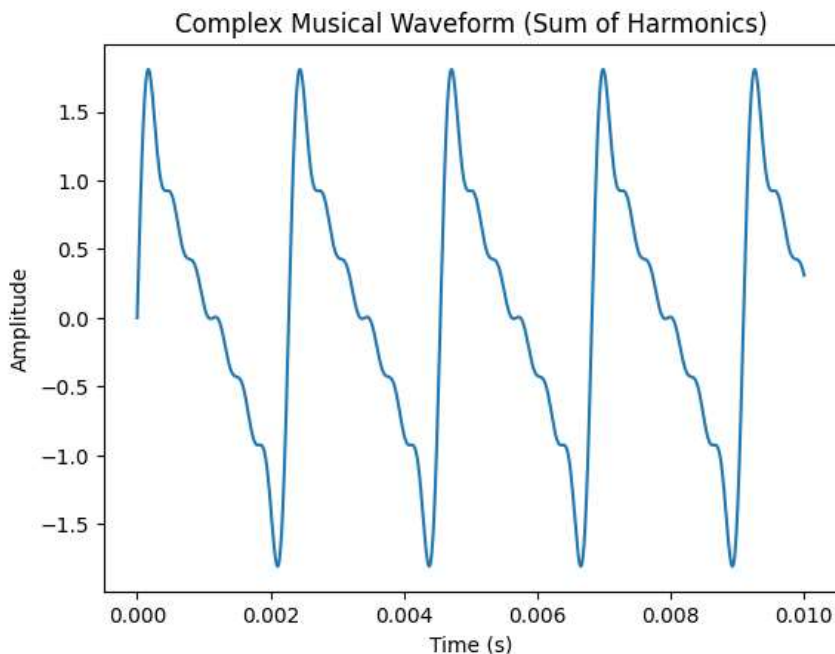
This means:

**A musical tone = sum of harmonics**

- ❑ Each **sinusoid** corresponds to a **harmonic** frequency
- ❑ This allows complex sounds to be analyzed mathematically

**Real instrument sounds are complex**

❑ **They are not pure sine waves, but they are still periodic**



**Typical for real musical instruments**

**Instrument**

**Flute**

**Violin**

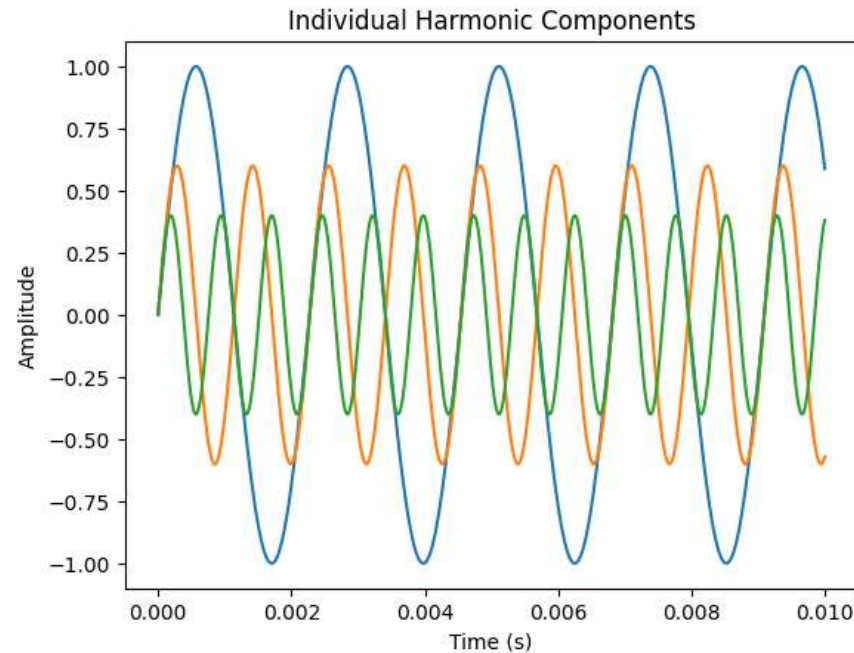
**Waveform type**

**Nearly sinusoidal**

**Rich harmonics**

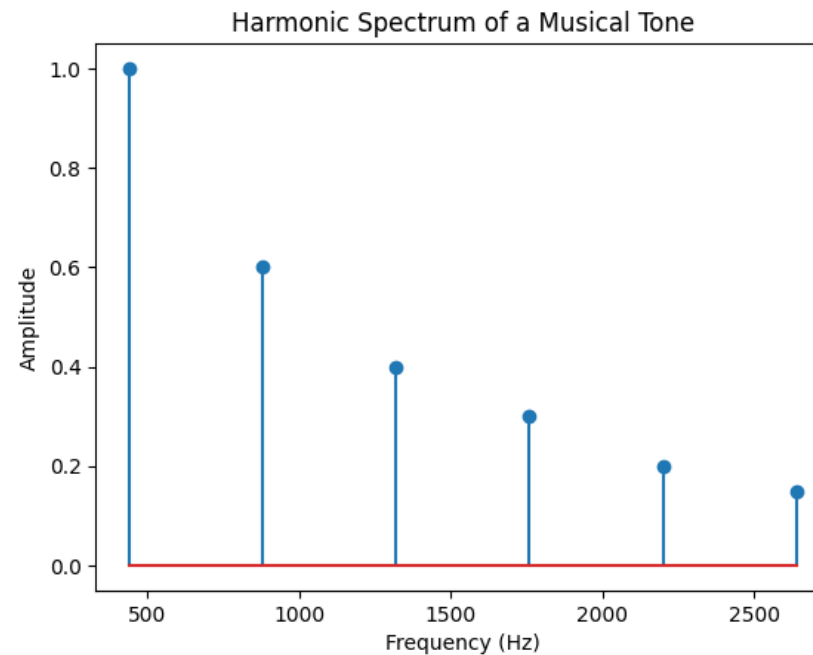
A musical tone consists of **a fundamental frequency** and **harmonics**

- ❑ Each harmonic is a multiple of the fundamental
- ❑ Different **amplitudes** create different **timbre**



**Fourier Series reveals frequency components**

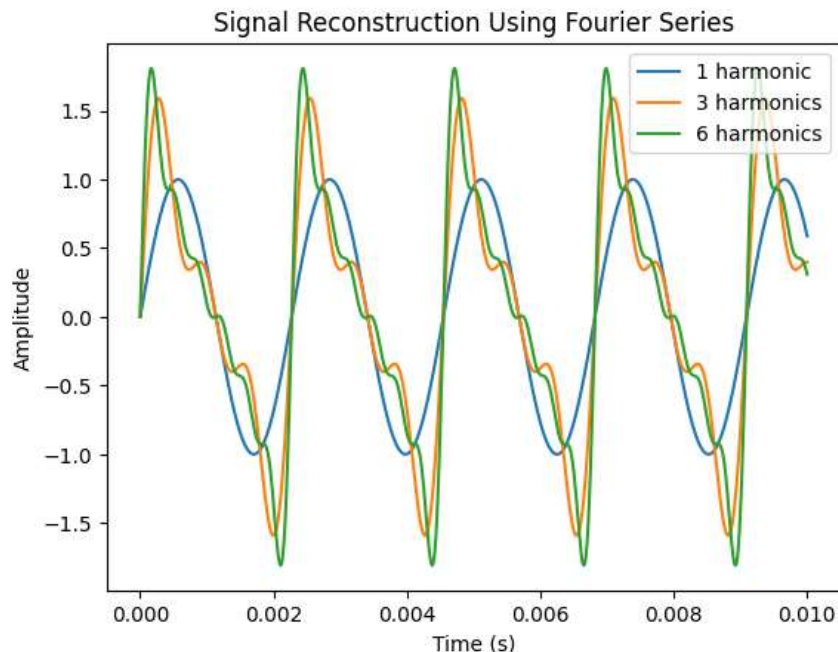
- ❑ The spectrum shows **amplitude vs frequency**
- ❑ Harmonics appear as **discrete spectral lines**



A complex sound can be reconstructed by adding harmonics

❑ Adding more harmonics improves accuracy

❑ This principle is used in **sound synthesis**



## Case 1: 1 harmonic

Only the fundamental frequency.

Result:

$$x(t) = A_1 \sin(\omega_0 t)$$

This sounds like a **pure tone** (tuning fork)

## Case 2: 3 harmonics

$$x(t) = A_1 \sin(\omega_0 t) + A_2 \sin(2\omega_0 t) + A_3 \sin(3\omega_0 t)$$

The waveform becomes more complex

The sound becomes **richer**

## Case 3: 6 harmonics

**More harmonics** create a **waveform closer to real** instrument sound

This is exactly how sound synthesis works

Electronic synthesizers generate tones by **combining harmonics**

- ❑ Audio compression analyzes frequency components
- ❑ Speech and music processing rely on spectral analysis





**Electronic instruments generate sounds using **Fourier synthesis****

## **Step 1 — Choose pitch**

**Select fundamental frequency**

**Example:**

$$f_0 = 440 \text{ Hz}$$

## **Step 2 — Choose harmonic amplitudes**

**Example:**

$$A_1 = 1, A_2 = 0.6, A_3 = 0.4, A_4 = 0.3$$

**These determine instrument character**

## **Step 3 — Generate waveform**

**Add sinusoids**

$$x(t) = \sum A_k \sin(k\omega_0 t)$$

## **Step 4 — Output sound**

**The resulting waveform is played through speakers**



## Digital synthesizers

- Examples include systems from the company Yamaha
- They generate tones using **harmonic synthesis**



# Fourier synthesis

